Method Name

Sunday, 15 June 2025 15:53

Query Methods

https://docs.spring.io/spring-data/jpa/reference/jpa/query-methods.html

- JPA can generate queries **automatically based on the method names** you define in a repository interface.
- This is one of the **most convenient features** because it allows you to write clean, self-explanatory code without dealing with any SQL or JPQL directly.

How It Works:

- **Method Naming Convention:** The method's name follows a particular pattern: findBy, readBy, or queryBy followed by the entity's properties. Spring parses this name, translates it into the corresponding SQL/JPQL, and executes the query.
- **Supported Keywords:** You can use keywords such as And, Or, Between, LessThan, Like, and more. This helps in creating queries for multiple conditions.

Example:

```
public interface UserRepository extends JpaRepository<User, Long> {
   // Find users by last name
   List<User> findByLastName(String lastName);
// Find users by age range
   List<User> findByAgeBetween(Integer startAge, Integer endAge);
// Find user by first name and last name
   Optional<User> findByFirstNameAndLastName(String firstName, String lastName);
}


@Entity
public class User {
   @Id
   @GeneratedValue(strategy = GenerationType.IDENTITY)
   private Long id;

   private String firstName;
   private String lastName;
   private Integer age;

   // getters and setters
}
```

Advantages:

- **Simplicity:** No need to write boilerplate code for common queries.
- **Maintainability:** The intent is expressed in the method name, making the code self-documenting.
- **Type-Safety:** Since the method signature is well-defined, you get compile-time checking.

Limitations:

- **Complex Queries:** When queries become more intricate (e.g., involving joins, grouping, or subqueries), method names can become unwieldy or may not support the required complexity.

Keyword	Sample
Distinct	findDistinctByLastnameAndFirstname
And	findByLastnameAndFirstname
Or	findByLastnameOrFirstname
Is, Equals	findByFirstname,findByFirstnameIs,findByFirstnameEquals
Between	findByStartDateBetween
LessThan	findByAgeLessThan
LessThanEqual	findByAgeLessThanEqual
GreaterThan	findByAgeGreaterThan
GreaterThanEqual	findByAgeGreaterThanEqual
After	findByStartDateAfter
Before	findByStartDateBefore
IsNull, Null	findByAge(Is)Null
IsNotNull, NotNull	findByAge(Is)NotNull
Like	findByFirstnameLike
NotLike	findByFirstnameNotLike
StartingWith	findByFirstnameStartingWith
EndingWith	findByFirstnameEndingWith
Containing	findByFirstnameContaining
OrderBy	findByAgeOrderByLastnameDesc
Not	findByLastnameNot
In	findByAgeIn(Collection<Age> ages)
NotIn	findByAgeNotIn(Collection<Age> ages)
True	findByActiveTrue()
False	findByActiveFalse()
IgnoreCase	findByFirstnameIgnoreCase